

Evaluation of Left Ventricular Metrics Using Echocardiography

BMEN 6003 – Module 2 Report

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Abstract—This report compares three approaches to estimate cardiac metrics from ultrasound data: geometric approximation from 2-D views, Doppler-based cardiac output, and a modified Simpson method implemented in Python. The workflow estimates end-diastolic and end-systolic ventricular volumes and derives stroke volume, ejection fraction, and cardiac output. The study also documents the software implementation used to perform frame selection, scale calibration, manual measurements, and final metric computation.

Index Terms—Echocardiography, left ventricle, stroke volume, ejection fraction, modified Simpson method, Doppler.

I. INTRODUCTION

Cardiovascular ultrasound is a standard noninvasive tool to evaluate ventricular morphology and function in clinical practice [1]. In this module, left-ventricular function was assessed from volunteer echocardiography data acquired during Week 2 and processed during Week 3.

The report addresses the metrics determination through three complementary approaches: (1) a geometric 2-D approximation of ventricular volume, (2) a Doppler-based estimate of cardiac output, and (3) a modified Simpson approach based on long-axis and short-axis views.

A. Background and Hypothesis

Single-view geometric methods are fast but sensitive to shape assumptions and manual measurement uncertainty. Doppler methods estimate flow directly but are sensitive to outflow tract diameter and insonation conditions. The modified Simpson method combines multiple short-axis measurements and is therefore expected to provide more robust EDV and ESV estimates for the same subject.

B. Objective

The objective is to compare EDV, ESV, SV, EF, and CO across the methods, identify the main sources of uncertainty, and document a reproducible Python implementation used to obtain these metrics.

II. MATERIALS AND METHODS

A. Data and Software Environment

The analysis uses one apical four-chamber video and three short-axis videos (MV, PM, AP), together with Doppler measurements and heart rate recorded during the laboratory sessions. Processing was implemented

in Python using a dedicated application (`main.py`, `heartvolume/gui/app.py`) and numerical routines in `heartvolume/core/calculations.py`.

B. Handout-Driven Analysis Logic

Following the Week 2&3 instructions, the report compares three cardiac-function techniques: (i) Week 2 Simple Method for EDV/ESV, (ii) Week 2 Doppler method for flow-derived CO, and (iii) Week 3 Modified Simpson method for EDV/ESV. For volume-based methods (Simple and Simpson), SV, EF, and CO are computed from EDV and ESV. Doppler CO is computed independently from VTI , HR , and D_{AO} .

C. Week 2: Simple Method (2-D Geometric Approximation)

The first ventricular-volume approach is referred to as the *Simple Method*. Assuming the ventricular cavity as a truncated axisymmetric ellipsoid, volume is estimated by

$$V = \frac{\pi}{6} D_{\text{endo}}^2 L_{\text{endo}}. \quad (1)$$

The same formula was applied at end-diastole and end-systole to obtain EDV_{simple} and ESV_{simple} , consistent with standard echocardiographic quantification practice [2].

D. Week 2: Balloon Phantom Formula

The balloon is modeled as an ellipsoid with major axis L and orthogonal minor axes D_1 and D_2 :

$$V_{\text{balloon}} = \frac{4}{3} \pi \left(\frac{L}{2} \right) \left(\frac{D_1}{2} \right) \left(\frac{D_2}{2} \right) = \frac{\pi}{6} L D_1 D_2. \quad (2)$$

In this report, four sets of measurements are provided. Each set is used to compute an estimated balloon volume, then absolute and relative errors versus the known injected volume.

E. Week 2: Doppler-Derived Cardiac Output

Cardiac output from aortic outflow was computed as

$$CO_{\text{Doppler}} = VTI \times HR \times \frac{\pi D_{AO}^2}{4}. \quad (3)$$

The aortic diameter was measured directly on the D_{AO} image/video after calibration, and VTI and HR were entered from the acquisition session.

F. Week 3: Modified Simpson Method Implemented in Python

The Python workflow reproduces the Week 3 procedure with interactive frame selection and calibrated manual measurements:

- frame extraction from the four synchronized views;
- automatic or manual scale calibration in cm/pixel;
- manual distance measurements on ED and ES frames;
- modified Simpson volume computation through `calculate_volume_simpson`.

For each cardiac phase, the implementation uses long-axis length L and the three short-axis diameters at apex, papillary muscle, and mitral valve levels. Using areas A_{AP} , A_{PM} , and A_{MV} ,

$$V_{\text{Simpson}} = \frac{L}{3} \times \frac{A_{AP} + A_{MV} + 4A_{PM}}{3}, \quad (4)$$

with

$$A_i = \pi \left(\frac{D_i}{2} \right)^2, \quad i \in \{AP, PM, MV\}. \quad (5)$$

This provides EDV_{simpson} and ESV_{simpson} .

G. Derived Metrics and Comparison Framework

For each volume-based method (Simple and Modified Simpson), stroke volume and ejection fraction were computed as

$$SV = EDV - ESV, \quad (6)$$

$$EF = \frac{SV}{EDV} \times 100, \quad (7)$$

$$CO = SV \times HR. \quad (8)$$

Thus, three cardiac output estimates are compared in this report:

- $CO_{\text{simple}} = SV_{\text{simple}} \times HR$,
- $CO_{\text{simpson}} = SV_{\text{simpson}} \times HR$,
- $CO_{\text{doppler}} = VTI \times HR \times (\pi D_{AO}^2 / 4)$.

The full comparison is therefore organized as follows:

- compare ventricular volumes from Simple Method vs Modified Simpson,
- compare cardiac output from Simple vs Modified Simpson vs Doppler,
- compare balloon estimated volume vs true balloon volume (250 mL).

In the Python code, `build_volume_result` and `build_doppler_result` centralize these calculations and standardize units (mL and L/min) for consistent reporting across methods.

H. Error Analysis Strategy

Method comparison focused on sensitivity to manual annotations, frame selection, calibration quality, and geometric assumptions. Differences between methods were interpreted by considering expected physiological ranges and known acquisition-related sources of bias.

III. RESULTS

A. Measurement Summary for the Simple Method

Primary measurements collected from ED and ES frames are summarized in Table I. Doppler inputs used for outflow-based cardiac output are listed below the table.

TABLE I
PRIMARY MEASUREMENTS EXTRACTED FROM ULTRASOUND IMAGES

Parameter	ED	ES
L_{endo} (cm)	7.01	5.66
D_{endo} (cm)	4.21	3.86

Doppler inputs: $VTI = 19$ (cm), $D_{AO} = 2.06$ (cm), and $HR = 74$ (bpm).

B. Supplementary Measurements for Modified Simpson

Detailed input dimensions used by the Modified Simpson method are reported in Table II. These values correspond to manual measurements performed on ED and ES frames for the four synchronized views.

TABLE II
MODIFIED SIMPSON INPUT MEASUREMENTS

Parameter	ED (cm)	ES (cm)
L (apical 4-chamber)	5.30	4.29
D_{MV} (mitral valve level)	4.77	2.07
D_{PM} (papillary muscle level)	3.00	1.72
D_{AP} (apex level)	2.48	1.63

C. Comparison of Cardiac Metrics Across Methods

Derived ventricular metrics are reported in Table III. The Simple Method and Modified Simpson provide EDV and ESV directly, whereas Doppler provides flow-derived CO. This table is used for the required comparisons: volumes (Simple vs Modified Simpson) and cardiac output (Simple vs Modified Simpson vs Doppler).

TABLE III
CARDIAC METRICS OBTAINED WITH EACH METHOD

Metric	Simple Method	Doppler	Modified Simpson
EDV (mL)	65.05	N/A	30.00
ESV (mL)	44.15	N/A	7.05
SV (mL)	20.91	N/A	22.95
EF (%)	32.14	N/A	76.5
CO (L/min)	1.58	4.67	1.70

D. Balloon Phantom Validation

Balloon phantom validation results are shown in Tables IV and V. Four sets of measurements are reported (including the required two) and compared with the known balloon volume ($V_{\text{true}} = 250$ mL) to quantify measurement error.

TABLE IV
BALLOON PHANTOM MEASURED DIMENSIONS USED FOR VOLUME ESTIMATION

Set	L (cm)	D_1 (cm)	D_2 (cm)
1	8.31	5.34	6.01
2	8.51	6.21	6.22
3	9.07	5.76	6.22
4	8.93	6.51	5.81

TABLE V
BALLOON PHANTOM VOLUME VALIDATION ACROSS FOUR MEASUREMENT SETS

Set	Estimated volume (mL)	Absolute error (mL)	Relative error (%)
1	140	110	44.0
2	172	78	31.2
3	170	80	32.0
4	177	73	29.2

E. Software Outputs

The Python application produces a complete run log for each method, including selected frame indices, calibrated scale values, measured distances, and final EDV/ESV/SV/EF/CO metrics. This output was used as the numerical source for all tables in this section.

F. Figures

Representative images associated with the workflow include frame selection, scale calibration, and final metric display for EDV and ESV analyses. These visual artifacts are part of the project assets and were used to verify measurement consistency across views.

IV. DISCUSSION

The evaluated approaches provide complementary information on left-ventricular function. The Simple Method offers a rapid estimate with minimal inputs, the Doppler method provides a direct flow-based perspective, and the Modified Simpson approach uses multi-view geometry to reduce dependence on a single diameter assumption.

A. Interpretation of Method Differences

The primary volume comparison in this report is $\{EDV, ESV\}_{\text{simple}}$ versus $\{EDV, ESV\}_{\text{simpson}}$. In practice, discrepancies are expected because the Simple Method relies on a stronger geometric assumption, while Modified Simpson integrates multiple short-axis measurements. Doppler-derived CO can remain physiologically coherent even when volume-based estimates differ, but it is strongly dependent on accurate D_{AO} measurement because cross-sectional area scales with diameter squared.

B. GitHub Repository and Auto-Detection Features

The full codebase and usage documentation are available in the repository. Beyond the core measurement workflow, several auto-detection features were tested in this project: automatic data-path discovery for the required videos/images,

automatic ruler scale detection on the right side of each frame with interactive ROI correction, and iterative automatic contour/ellipse tracking strategies for ultrasound sequences. These developments were used to evaluate how far automation can support or accelerate manual echocardiographic analysis. Additional implementation details, visual examples, and tested variants are provided in the [repository](#).

C. Experimental Automatic Contour Tracking

Beyond the semi-automated workflow, several fully automatic contour tracking methods were explored to segment the left ventricle. These features remain experimental as they did not yield results robust enough for clinical use, but they highlight common challenges in echocardiographic image processing. Three main approaches were investigated:

- 1) **Threshold-based Region Growing & Ellipse Fitting:** This classical computer vision method starts from a user-drawn contour on the first frame. An algorithm then iteratively expands the region by analyzing brightness differences between the inner cavity and an outer ring, aiming to detect the endocardium wall. An ellipse is fitted to the resulting contour. On subsequent frames, the ellipse from the previous frame is used as a seed. While functional on high-contrast images, this method struggles significantly with the speckle noise and signal dropouts characteristic of clinical ultrasound data.
- 2) **Optical Flow (Lucas-Kanade):** This approach attempted to track keypoints along the manually-traced endocardium border across frames. Using the Lucas-Kanade optical flow algorithm, the displacement of these points was calculated. However, this method proved unreliable. The constantly changing speckle pattern (speckle noise decorrelation) of the myocardial tissue causes the tracked points to drift away from the anatomical border rapidly.
- 3) **Deep Learning (EchoNet-Dynamic):** A state-of-the-art deep learning model, EchoNet-Dynamic, was also evaluated. This model is a segmentation architecture specifically trained on thousands of apical four-chamber echocardiograms. Surprisingly, this approach was the least conclusive on the available dataset, which may be attributed to the low quality of the input videos compared to the model's training data.

These experiments confirm that robust, automated segmentation of the left ventricle in noisy ultrasound sequences is a complex task that typically requires more advanced, dedicated models and high-quality image acquisition. More details can be found in the [github repository](#).

D. Main Sources of Uncertainty

The dominant uncertainty sources were frame timing (true ED and ES), calibration quality, manual point placement, and 2-D out-of-plane effects. For Doppler, uncertainty in aortic diameter and alignment of the velocity profile remained the two main contributors to CO variability.

E. Comparison With Expected Physiological Ranges

The final metrics were interpreted against expected resting ranges for healthy adults. The Doppler-based cardiac output (CO) aligns well with typical physiological expectations. In contrast, the CO values derived from both the Simple Method and the Modified Simpson method are approximately half of the expected value, although they are consistent with each other.

This large discrepancy likely originates from the manual annotation process. Accurate placement of measurement points on echocardiographic images is a challenging task that requires significant clinical expertise to correctly identify anatomical landmarks, especially in the presence of image noise and artifacts. As a novice, it is highly probable that my manual measurements for ventricular diameters and lengths were systematically underestimated, leading to erroneously low volume calculations and, consequently, a cardiac output that is not representative of the true physiological state. This highlights the critical role of operator experience in achieving reliable quantitative results from echocardiography.

F. Conclusion

This study compares echocardiographic quantification strategies while documenting a full computational workflow. The Python implementation provides a practical and reproducible alternative to Matlab for this assignment while preserving methodological equivalence with the handout requirements. For this dataset, Modified Simpson is retained as the primary volumetric reference, the Simple Method serves as a rapid baseline, and Doppler provides a complementary flow-based validation through an independent CO estimate.

APPENDIX A

APPENDIX: PYTHON IMPLEMENTATION DETAILS

The analysis was implemented as a Python desktop application with a GUI entry point in `main.py` and the main workflow class in `heartvolume/gui/app.py`.

A. Program Structure

The implementation is organized into three layers:

- user interface and workflow orchestration (`HeartVolumeApp`),
- image/video interaction tools (frame selection, scale detection, distance measurement),
- cardiac metric computations in `heartvolume/core/calculations.py`.

B. Core Computation Functions

```
def calculate_volume_simple(L_cm, D_cm):  
    return (pi / 6.0) * D_cm**2 * L_cm  
  
def calculate_volume_simpson(L_cm, D_apex_cm, D_pm_cm, D_mv_cm):  
    A_apex = pi * (D_apex_cm / 2.0)**2  
    A_pm   = pi * (D_pm_cm   / 2.0)**2  
    A_mv   = pi * (D_mv_cm   / 2.0)**2  
    return (L_cm / 3.0) * (A_apex + A_mv + 4.0 * A_pm) / 3.0
```

```
def build_volume_result(EDV, ESV, HR=None):  
    SV = EDV - ESV  
    EF = 100.0 * SV / EDV  
    CO = (SV * HR / 1000.0) if HR is not None else  
    return EDV, ESV, SV, EF, CO
```

C. Execution Workflow

The software exposes three run modes corresponding to the report methods:

- `run_simple` for geometric EDV/ESV estimation,
- `run_simpson` for 4-view modified Simpson analysis,
- `run_doppler` for outflow-based SV and CO estimation.

Each run logs frame indices, calibrated scale values, measured distances, and final cardiac metrics used in the Results section.

D. Reproducibility Notes

To reproduce the computations, the final submission archives the exact input videos/images, selected ED/ES frames, calibrated cm/pixel values, and the console log generated by the application for each method.

REFERENCES

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